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Considerations in Living Kidney Donation

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CHAPTER OUTLINE

KEY POINTS

- The donor candidate evaluation should be completed efficiently but also comprehensively to meet the needs of the donor candidate, intended recipient, and health care system.
- Donor candidates should be provided with individualized estimates of short- and long-term risks, and risks should be considered with respect to predetermined program acceptance thresholds.
- Explicit recognition of perspectives of comparison is critical for drawing inferences about donor health outcomes (e.g., estimation of predonation risk, absolute postdonation risk, and donation-attributable risk).
- Recent studies grounded in comparison to healthy nondonor controls have identified small, donation-attributable increases in the long-term risks of kidney failure and gestational hypertension and preeclampsia in living kidney donors. This information should be shared with donor candidates.
- Simultaneous consideration of each donor candidate's profile of demographic and health characteristics advances a new framework for consistent and transparent risk assessment. New risk prediction tools are available to help apply this framework in practice.
- Accurate assessment of kidney function is central to the donor candidate evaluation and may be efficiently assessed by estimated glomerular filtration rate (GFR) screening, followed by confirmatory testing (e.g., measured GFR or creatinine clearance).
- If a donor candidate is of sub-Saharan African ancestry, testing for *APOL1* renal risk variants may be offered, and associated costs should be covered as part of the candidate evaluation. Candidates should be informed that the presence of two *APOL1* renal risk variants increases the lifetime chance of developing kidney failure even in the absence of donation; however, the effects of kidney donation on this risk are not currently known.
- Local laws and regulations on living donation must be followed and explained to donor candidates, and donor autonomy should be respected during all phases of evaluation and donation.

INTRODUCTION

The first successful living donor kidney transplantation (LDKT) occurred on December 23, 1954, between identical twins, Ronald (donor) and Richard (recipient) Herrick.[1,2] The operation was performed by Dr. Joseph Murray at the Peter Bent Brigham Hospital in Boston and was recognized by the Nobel Prize in Medicine in 1990. Over the ensuing 60 years, advances in immunosuppression expanded opportunities for living donations from identical twins to other biologic relatives and then to unrelated persons with essentially equivalent outcomes, regardless of the donor-recipient relationship.[3,4] Opportunities for successful LDKT have expanded further over recent decades through the widespread access to laparoscopic donor nephrectomy, acceptance of donor candidates with medical complexities such as predonation hypertension, and the use of techniques to overcome immune incompatibility, such as kidney paired donation (KPD) programs.[5]

Worldwide, approximately 27,000 LDKT are performed each year.[6] Rates of LDKT vary across nations with the highest rates, per million population (pmp), in countries such as Saudi Arabia (32 pmp), Jordan (29 pmp), Iceland (26 pmp), Iran (23 pmp), and the United States (21 pmp). In many regions, the LDKT rates have increased over the last decade, with 62% of countries reporting at least a 50% increase, whereas in many other regions, rates have become stagnant or have steadily declined. Generally, LDKT represent almost 40% of all kidney transplantations performed; however, in some countries, such as Pakistan, India, Iran, and Japan, LDKT comprise over 80% of all kidney transplantations. Thus, considerations for the safe, effective, and ethical practices in living kidney donation have global significance, affecting over 90 countries worldwide.

CONSIDERATIONS FOR THE KIDNEY TRANSPLANT RECIPIENT

For eligible patients with end-stage kidney disease (ESKD), kidney transplantation is the preferred treatment option compared with chronic dialysis because it is associated with superior long-term patient survival, improved quality of life, and lower health care costs.[7–10] Unfortunately, there is a shortage of organs to meet the need, resulting in a high proportion of patients dying while on the kidney transplant waiting list.[4] LDKT is a key strategy to help meet the growing demand for kidney transplants.

Compared with deceased donor kidney transplantation (DDKT), LDKT is also associated with superior patient and graft survival.[11] The 5-year, age-adjusted patient survival is higher for LDKT recipients (94%) than DDKT recipients (76%), and both are higher than remaining on the kidney transplant waiting list (60%).[11] Graft survival at 1, 5, and 10 years is also higher after LDKT versus DDKT (10-year graft survival, 63% vs. 47%).[4,12]

Other benefits for the recipient include the potential for preemptive transplantation or shorter waiting time until transplantation, allowing recipients to avoid or decrease the risks associated with chronic dialysis.[13,14] Furthermore, LDKT can be scheduled with advance notice at a time that works well for donors and recipients, compared with the urgent nature of DDKT. This may allow for better optimization of pretransplantation conditions prior to surgery. LDKT may provide additional opportunities for better genetic matching and a lower risk of rejection. Finally, with LDKT, the cold ischemia time is shortened, and the risk of delayed graft function is decreased. For all of these reasons, LDKT offers advantages to the recipient compared with DDKT.

Based on clear evidence of recipient benefit and surveys supporting positive public perceptions of living donation, a number of initiatives are underway to increase awareness of LDKT and reduce disincentives.[15] A 2015 American Society of Transplantation (AST) Live Donor Community of Practice consensus conference recognized LDKT as the “best treatment option” for eligible patients with kidney failure.[16] The importance of educating and supporting patients, families, and front-line renal health care providers to enable informed decisions about kidney transplantation and living kidney donation has been receiving greater attention. For example, education tools have been developed to provide individualized, risk-adjusted outcomes to help transplant candidates understand the relative benefits of LDKT and various types of DDKT compared with chronic dialysis.[17,18] Also, there have been greater efforts to support kidney patients finding living kidney donors.

EVALUATION AND SELECTION OF THE LIVING KIDNEY DONOR CANDIDATE

The practice of LDKT is based on the principle that the benefits to the recipient (see “Considerations for the Kidney Transplant Recipient”) outweigh the minimal risks to the carefully evaluated and selected living donor (see “Risks and Outcomes of Living Kidney Donation”). Living kidney donors should undergo a rigorous evaluation and selection process to ensure that the short- and long-term risks to the donor are minimized (Fig. 71.1). In addition to this, the benefits and risks to the intended recipient are also considered. From the recipient’s perspective, an aim is to select donors who will provide adequate graft function while minimizing the transmission of any donor-derived diseases, such as infections or malignancy. To mitigate potential conflict of interest, it is recommended that the evaluations of the donor candidate and intended recipient be performed by separate, independent teams.[19–23] Moreover, the living donor evaluation process should be completed in a timely fashion to improve the donor candidate experience and optimize recipient outcomes.[13,14,24] One international multicenter study has reported that the median time from the beginning of the donor candidate evaluation until donor nephrectomy is 10 months.[24] Avoidable delays can lead to dialysis initiation for preemptive transplant candidates, lengthen dialysis time for patients on dialysis, and reduce donor satisfaction.[13,25] Opportunities to improve efficiency include the choice, timing, and sequence of evaluation tests and visits, use of navigators, and monitoring of evaluation timeliness as a quality metric.[26]

Clinical Relevance

An efficient living donor candidate evaluation is completed in as little time as possible and meets the needs of the donor candidate, intended recipient, and health care system. Avoidable delays can lead to dialysis initiation for preemptive transplant candidates, lengthen dialysis time for patients on dialysis, and reduce donor satisfaction. Opportunities to improve efficiency include the choice, timing, sequencing of evaluation tests, procedures and consults, use of navigators, and monitoring of evaluation timeliness as a quality metric.

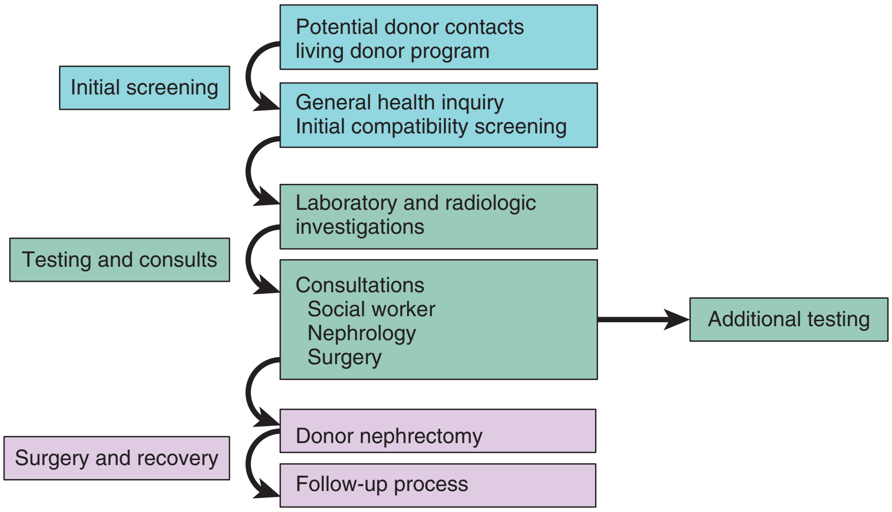

Fig. 71.1 Overview of the living kidney donor evaluation process.

Table 71.1 Existing Guidelines and Policies on the Evaluation of Living Kidney Donors

Guidelines	Year	National/ International
Kidney Disease: Improving Global Outcomes (KDIGO): Evaluation and Care of Living Kidney Donors[30]	2017	International
Organ Procurement and Transplantation Network (OPTN) Policy 14: Living Donation[23]	2013 2015 (update)	United States
Kidney Paired Donation (KPD) Protocol for Participating Donors[137]	2015	Canada
European Renal Best Practice (ERBP) Guideline on Kidney Donor and Recipient Evaluation and Perioperative Care[22]	2013	Europe
British Transplantation Society (BTS): United Kingdom Guidelines for Living Donor Transplantation[21]	2011	United Kingdom
Caring for Australasians with Renal Impairment (CARI)[138]	2010	Australia/New Zealand
A Report of the Amsterdam Forum on the Care of the Live Kidney Donor: Data and Medical Guidelines[139]	2005	International

Multiple guidelines have assisted clinicians in the complex process of donor evaluation and selection (Table 71.1). A systematic review of these clinical practice guidelines has found that although many recommendations were consistent, important variations exist, and many appeared to lack methodologic rigor.[27] The 2017 Kidney Disease: Improving Global Outcomes (KDIGO) "Clinical Practice Guideline on the Evaluation and Care of Living Kidney Donors" provides a comprehensive set of best practice recommendations based on a systematic evidence review, de novo evidence generation, and expert opinion when evidence is lacking.[19] When possible, the guideline recommends that transplantation programs establish numeric thresholds for short- and long-term post-donation risks, above which the program will not accept the candidate for donation (Fig. 71.2). It also demonstrates how tools can be developed to help estimate a donor candidate's risk of long-term complications such as ESKD based on their individualized set of predonation demographic and health characteristics.

A central goal of the KDIGO guideline is to promote "consistent, transparent and defensible decision-making" based on comparisons of individualized, quantitative estimates of donor risks "to a transplant program's acceptable risk threshold."[19] Risk threshold is defined as the upper limit of acceptable risk established by a program for donor candidate selection. Under this framework, when a candidate's estimated risk is above the acceptable threshold, the transplant program is justified in declining the candidate and can ground its decision in a quantitative framework. When a donor candidate's estimated risk is below the acceptable risk threshold, the transplant program should accept a donor candidate, and it should be the candidate's decision whether to proceed with living kidney donation after being informed of the risks. Once established, acceptable risk thresholds should be applied consistently and transparently for all donor candidates evaluated at a program. The KDIGO framework was informed by a systematic evidence review.[28] The KDIGO group also developed a tool to quantify a donor candidate's risk of postdonation complications such as ESKD. This tool projects the 15-year and lifetime risk of ESKD based on the level of the predonation glomerular filtration rate (GFR) and other baseline demographic and health factors.[29] For practical applications, the resulting risk models were incorporated into an online risk prediction tool (http://www.transplantmodels.com/esrdrisk). This tool serves as an example and can be improved with future research efforts for various types of living kidney donors worldwide.

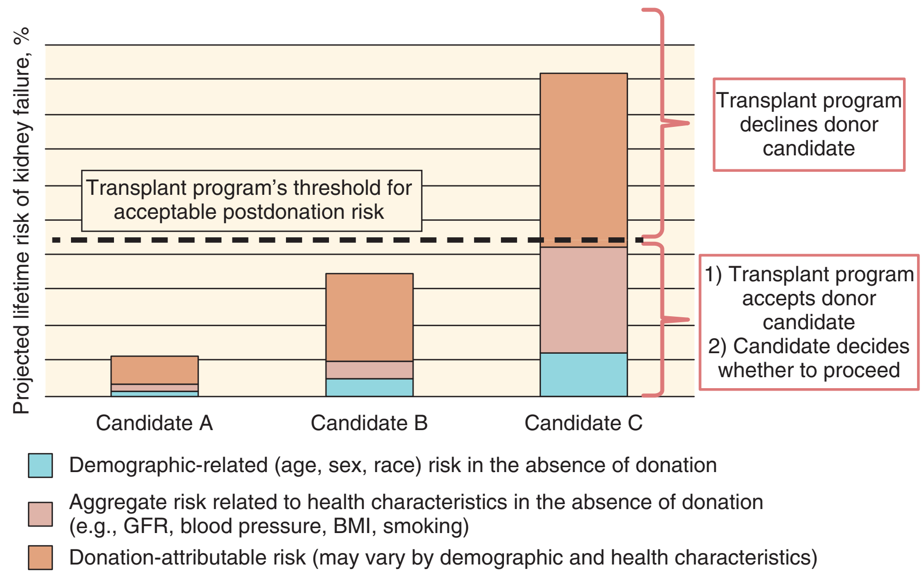

Fig. 71.2 Kidney Disease: Improving Global Outcomes framework to accept or decline donor candidates based on a transplantation program's threshold of acceptable projected lifetime risk of kidney failure, quantified as the aggregate of risk related to demographic and health profile and donation-attributable risks. *BMI,* Body mass index; *GFR,* glomerular filtration rate. (From Lentine KL, Kasiske BL, Levey AS, et al. KDIGO clinical practice guideline on the evaluation and care of living kidney donors. *Transplantation.* 2017;101(8 Suppl 1):S7–S105.)

Clinical Relevance

Simultaneous consideration of each donor candidate's profile of demographic and health characteristics advances a new framework for consistent and transparent risk assessment. As demonstrations of proof of concept, online tools have been developed to estimate the risk of kidney failure in healthy persons based on 10 demographic and health traits (currently in the absence of donation), as well as postdonation risk based on a more limited set of factors. These tools can be refined and expanded with more precise risk information (e.g., risk based on longer follow-up, additional risk factors such as emerging genetic markers, and integration of predonation and postdonation risks). Estimation of risk using the best available data, and communication of risks and associated uncertainty in a manner easily understood by donor candidates, can support evidence-based donor selection and shared decision making.

The evaluation process should include a comprehensive history, physical examination, laboratory and radiologic investigations, and specialist consultations (Table 71.2). Aspects of the process may vary by region and transplantation center, including the order and timing of the components and what is considered required or additional testing. Depending on local resources and policies, transplantation programs may also choose to evaluate multiple donors for an intended recipient, either simultaneously or sequentially. The 2017 KDIGO living kidney donor guideline recommends that all donor candidates should be evaluated using the same criteria, regardless of the intended recipient.[19] Table 71.3 summarizes several contraindications to living kidney donation.

KIDNEY FUNCTION

The purpose of evaluating GFR in kidney donor candidates is to detect kidney disease and project long-term outcomes for the candidate and the recipient should they go on to donation. Recommended methods for evaluating GFR in donor candidates are based on the 2012 KDIGO Chronic Kidney Disease (CKD) guideline.[30,31] Considering practicality, test availability, and costs, the 2017 KDIGO living donor guideline recommends an initial estimated GFR (eGFR) based on the serum creatinine level (eGFRcr) and confirmation using one or more of the following measurements according to their availability: measured GFR (mGFR) from clearance of exogenous radiolabeled filtration markers, measured creatinine clearance (mCrCl) based on collecting a timed (24-hour) urine specimen, eGFR based on serum creatinine and cystatin levels (eGFRcr-cys), or repeated eGFRcr, with the latter being the least preferred approach.[19,30,31] Although mGFR or mCrCl is required for donor evaluation in the United States according to Organ Procurement and Transplantation Network (OPTN) policy, a timed urine collection for the albumin excretion rate (AER) is not required—measurement of urine protein or albumin may be performed on a random "spot" urine sample. In countries where clearances are required for the assessment of GFR, an efficient strategy may be to omit timed urine collections and rely on the mGFR using clearance of an exogenous filtration marker and a random urine albumin-to-creatinine ratio (ACR). In countries where clearance measures are not required for the assessment of GFR, transplantation programs could obtain the eGFRcr, eGFRcr-cys, and urine ACR prior to a donor candidate's visit to the center (Fig. 71.3).[32]

An Internet-based application has been published to compute the probability of mGFR above or below clinically-appropriate thresholds based on the results of eGFR testing

Table 71.2 Components of the Living Kidney Donor Evaluation

Medical and Surgical History	Examples
Disease or Condition	
Genitourinary disease	Hematuria Proteinuria Kidney stones
Cardiac disease	Coronary artery disease Myocardial infarction Heart failure Arrhythmia
Cardiac risk factors	Hypertension Impaired fasting glucose (IFG) Impaired glucose tolerance (IGT) Diabetes mellitus Hyperlipidemia Exercise tolerance
Peripheral vascular disease	Intermittent claudication
Pulmonary disease	Asthma Chronic obstructive pulmonary disease Obstructive sleep apnea
Gastrointestinal disease	Peptic ulcer disease Inflammatory bowel disease
Hematologic disease	Deep vein thrombosis Pulmonary embolism Bleeding disorders Blood transfusions
Neurologic disease	Stroke Parkinson disease Neurodegenerative disease
Autoimmune disease	Systemic lupus erythematosus Rheumatoid arthritis
Rheumatologic disease	Gout
Psychiatric conditions	Anxiety Depression Suicide attempts
Infections	Urinary tract infection Hepatitis B and C viruses (HBV, HCV) Human immunodeficiency virus (HIV) Syphilis Sexually transmitted infections Severe acute respiratory syndrome Meningitis, encephalitis Creutzfeldt-Jakob disease Tuberculosis
Cancer	Melanoma Renal cell carcinoma Breast cancer Lung cancer Colon cancer Hematologic cancer (e.g., leukemia, lymphoma) Monoclonal gammopathy
Pregnancy-related complications	Gestational hypertension Gestational diabetes Preeclampsia, eclampsia Future childbearing plans
Allergies and Medications	
Allergies	Anesthesia reactions
Nephrotoxic medications	Lithium Nonsteroidal antiinflammatory drugs (NSAIDs)
Chronic pain medications	Opioids
Family History	
Genitourinary disease	Alport syndrome Autosomal-dominant polycystic kidney disease (ADPKD) Kidney stones
Cardiac disease	Sudden cardiac death Early coronary artery disease
Cardiac risk factors	Hypertension Diabetes mellitus
Cancer	Renal cell carcinoma Breast cancer Colon cancer
Social History	
Smoking, alcohol, drug use	Abuse, dependency Intravenous drug use
Employment	Healthcare workers Veterinarians
Health insurance status	
Social supports	Marital status Living arrangements
Exposure history	Sexual history Tattoos Body piercings Travel or prolonged residency Country of origin, birth Incarceration
Physical Examination	
Vital signs	Blood pressure Heart rate and rhythm Body mass index (BMI) based on height and weight
Head and neck	
Cardiac examination	
Respiratory examination	
Abdominal examination	Prior surgeries, hernias, vascular bruits
Skin examination	Lymphadenopathy Mucocutaneous lesions Needle tracks
Laboratory Studies	
Compatibility	ABO blood group (including subtype A1, if indicated) Human leukocyte antigen (HLA) typing Crossmatch

Table 71.2 Components of the Living Kidney Donor Evaluation (Cont'd)

Medical and Surgical History	Examples
Renal	Serum creatinine level with estimation of glomerular filtration rate (eGFR)
	Urinalysis
	Urine microscopy
	24-hour urine (creatinine clearance, protein excretion, metabolic stone panel,[a] including calcium, oxalate, uric acid, citric acid, sodium)
	Protein-to-creatinine ratio (PCR)
	Albumin-to-creatinine ratio (ACR)
Blood work	Complete blood count (hemoglobin, white blood cells, platelets)
	International normalized ratio (INR)
	Partial thromboplastin time (PTT)
	Electrolytes (sodium, potassium, bicarbonate, calcium, phosphate, magnesium)
	Fasting lipid profile (total cholesterol, high-density lipoprotein, low-density lipoprotein, triglycerides)
	Pancreas enzymes (lipase, amylase)
	Liver enzymes (aspartate aminotransferase, alanine aminotransferase, alkaline phosphatase, gamma-glutamyl transferase)
	Bilirubin
	Albumin
	Uric acid
	Thyroid function test[a]
	Beta-human chorionic gonadotropin (β-hCG)[a]
Genetic renal disease[a]	Apolipoprotein L1 (*APOL1*) genotyping
	ADPKD mutation analysis
Diabetes mellitus	Fasting blood glucose
	Hemoglobin A1C
	Oral glucose tolerance test (OGTT)[a]
Infection	Urine culture and sensitivity
	Cytomegalovirus (CMV)
	Epstein-Barr virus (EBV)
	Hepatitis B (HBV surface antigen, HBV core antibody)
	Hepatitis C (anti-HCV)
	Human immunodeficiency virus (anti-HIV)
	Human T-lymphotrophic virus (HTLV)
	Herpes simplex virus (HSV)
	Varicella zoster virus (VZV)
	Treponema pallidum (syphilis)
	Mycobacterium tuberculosis (purified protein derivative, interferon-gamma release assay)
	Malaria[a]
	Trypanosoma cruzi[a]
	Schistosomiasis[a]
	Human herpes virus (HHV8)[a]
	Strongyloides[a]
	Typhoid[a]
	Brucellosis[a]
	West Nile virus[a]
	Zika virus[a]
	Chagas disease[a]
	Toxoplasmosis[a]
Cancer screening	Urine cytology
	Fecal occult blood test (FOBT)
	Fecal immunochemical test (FIT)[a]
	Colonoscopy[a]
	Prostate-specific antigen (PSA)[a]
	Pap smear[a]
Radiologic Studies	
Renal system	Abdominal computed tomography (CTA) or magnetic resonance angiography (MRA)
	Measured glomerular filtration rate (mGFR: iothalamate, ethylenediaminetetraacetic acid [EDTA], diethylenetriaminepentaacetic acid [DTPA])
	Cystoscopy[a]
	Renal biopsy[a]
Cardiac system	Electrocardiography
	24-hour Holter monitor[a]
	24-hour ambulatory blood pressure monitor (ABPM)[a]
	Echocardiography[a]
	Non-invasive myocardial perfusion study (exercise or pharmacological stress)
	Coronary angiography[a]
Pulmonary system	Chest x-ray
	Pulmonary function test (PFT)[a]
Cancer screening	Mammography[a]
	Low-dose chest computed tomography (CT)[a]
Specialist Consultation	
Psychosocial specialist	Social worker
	Psychologist
	Psychiatrist
Nephrologist	
Surgeon	
Other specialists[a]	Cardiology
	Infectious disease
	Dietitian

[a]If indicated, based on the donor demographic, history, physical examination, and/or laboratory or radiologic studies.

Table 71.3 Contraindications to Living Kidney Donation

Absolute Contraindications	Relative Contraindications
• Both <18 years old and mentally incapable of making an informed decision • High suspicion of donor coercion • High suspicion of illegal financial exchange between donor and recipient • ABO or HLA incompatibility, without a planned management protocol • Impaired kidney function (e.g., eGFR <60 mL/min/1.73 m^2) • Proteinuria (>300 mg/day) • Diagnosis of IgA nephropathy or Alport syndrome • Bilateral, recurrent kidney stones • Uncontrolled hypertension or hypertension with evidence of end-organ damage • Type 1 diabetes mellitus • Acute symptomatic infection (until resolved) • Chronic infection, without a management plan for donor and recipient • Active cancer or incompletely treated cancer • History of melanoma, testicular cancer, choriocarcinoma, hematologic cancer, monoclonal gammopathy, bronchial cancer, metastatic cancer • Current pregnancy • Uncontrolled psychiatric illness	• Two *APOL1* renal risk variants • Chronic illness (cardiac, pulmonary, gastrointestinal, neurologic) • Type 2 diabetes mellitus • Morbid obesity (e.g., BMI >35 mg/m^2) • Active substance use disorder

Modified from Lentine KL, Kasiske BL, Levey AS, et al. KDIGO clinical practice guideline on the evaluation and care of living kidney donors. *Transplantation*. 2017;101(8S Suppl 1):S7–S105; and Organ Procurement and Transplantation/United Network for Organ Sharing (OPTN/UNOS). Policies. https://optn.transplant.hrsa.gov/media/1200/optn_policies.pdf#nameddest=Policy_14.

(http://ckdepi.org/equations/donor-candidate-gfr-calculator). The algorithm was based on demographic characteristics obtained from the National Health and Nutrition Examination Survey (NHANES) and the test performance of eGFR (categoric likelihood ratios) from the CKD Epidemiology Collaboration.[33] The algorithm can be applied to living donor candidates and has been tested in a French cohort of 311 living donor candidates.[34] If there is a high probability that the mGFR is greater than a threshold needed for donation, and if the urine ACR is low, these tests could be repeated for confirmation without mGFR, mCrCl, or timed AER.[35] Alternatively, donor candidates with a high probability of the mGFR less than 60 mL/min/1.73 m^2 or high urine ACR could be reliably excluded from donation, with any additional testing for the purposes of caring for the candidate's long-term health. Donor candidates with the eGFR or urine ACR in intermediate ranges would require confirmatory tests with mGFR, mCrCl, and/or urine AER. Determining the eGFR and urine ACR in advance of a visit to the transplantation center allows planning for confirmatory tests with fewer visits to the center.

The physiologic renal adaptive response to nephrectomy in the donor is reviewed elsewhere.[36] Nephrectomy in a healthy donor (removal of 50% of the nephron mass) is associated with a fairly consistent adaptive increase of the remaining kidney's GFR such that the postdonation GFR is around 60% to 70% of the predonation value.[36] This information is relevant in considering an acceptable baseline predonation GFR. The 2017 KDIGO living kidney donor guideline recommends a GFR of 90 mL/min/1.73 m^2 or higher as an acceptable level of kidney function for donation; donor candidates with a GFR less than 60 mL/min/1.73 m^2 should not donate. The decision to approve donor candidates with a GFR of 60 to 89 mL/min/1.73 m^2 should be individualized based on their demographic and health profile (see Fig. 71.3).

ALBUMINURIA

Elevated protein in the urine (proteinuria) may suggest the presence or risk of developing kidney disease due to increased permeability of the glomeruli to protein and/or an inability of the renal tubules to reabsorb protein. Until acceptable standardization methods are available for quantifying deficiencies in tubular reabsorption, the urine albumin level remains the most reliable indicator of kidney disease, standardized to urinary creatinine as the ACR. The 2017 KDIGO living kidney donor guideline recommends an initial evaluation using the urine ACR in a random urine specimen, with confirmation by AER (from a timed urine specimen) or otherwise a second random urine ACR. Donor candidates with an AER more than 100 mg/day (or ACR >30 mg/mmol) should not donate. Such candidates have microalbuminuria and are at an elevated risk of developing CKD in their lifetime.[37] Candidates with an AER less than 30 mg/day (or ACR <3 mg/mmol or below the detectable limit of the assay) may be acceptable for donation; the decision to approve donor candidates with an AER of 30 to 100 mg/day should be individualized based on their demographic and health profiles in relationship to the transplantation program's acceptable risk threshold.

HEMATURIA

Persistent blood in the urine (hematuria) is another indicator for the presence or risk of developing kidney disease. The presence of hematuria is established by visualizing 2 to 5 red blood cells/high-powered field on microscopic evaluation. "Persistence" is established if hematuria is observed in more than 50% of urine samples obtained on two or three occasions. When hematuria is persistent, further investigation is warranted, which many include urine culture for

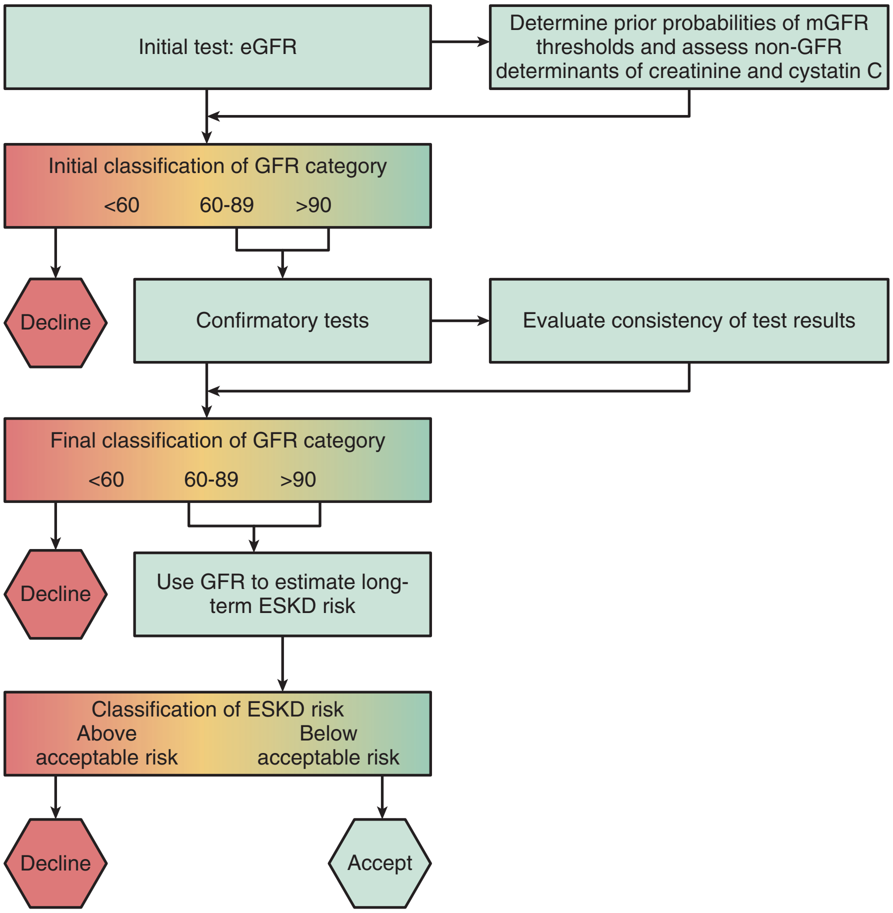

Fig. 71.3 Assessment of glomerular filtration rate. *Initial test,* eGFRcr is the initial test in most candidates. eGFRcys may be the preferred initial test for candidates with variations in non-GFR determinants of serum creatinine—for example, variation in muscle mass or diet. Interpretation of eGFR should include consideration of the probability that mGFR is above or below the threshold for decision making (http://ckdepi.org/equations/donor-candidate-gfr-calculator). A high likelihood that eGFR less than 60 mL/min/1.73 m^2 is justification for a decision to decline without further consideration. *Confirmatory tests,* mGFR or mCrCl is required in the United States. Elsewhere, eGFRcr-cys can be acceptable if mGFR or mCrCl is not available and eGFRcys was not used as the initial test. Repeat eGFRcr can be acceptable if none of the other confirmatory tests are available, but is not preferred. Inconsistent test results suggest inaccuracy of one or more tests, which should be discarded or repeated. Very high likelihood that mGFR less than 60 mL/min/1.73 m^2 is justification for a decision to decline without further consideration. *Use GFR to estimate long-term ESKD risk,* Long-term estimated risk of ESKD is compared with the transplantation center's threshold for acceptable risk. Long-term risk in the absence of donation can be computed from demographic and clinical characteristics, including the GFR (http://www.transplantmodels.com/esrdrisk). Additional risk attributable to donation is likely to be 3.5 to 5.2 times higher than risk in the absence of donation, but there is substantial uncertainty, especially in younger donor candidates, and we suggest caution in decision making. Postdonation risk above the threshold is justification for a decision to decline. Candidates with risk below the threshold are acceptable to make their own decision about whether to donate. Colors are blended together to signify that the threshold for decision making is imprecise. *eGFR,* estimated glomerular filtration rate; *eGFRcr,* estimated glomerular filtration rate based on serum creatinine; *eGFRcr-cys,* estimated glomerular filtration rate based on serum creatinine and cystatin C; *eGFRcys,* estimated glomerular filtration rate based on cystatin C; *ESKD,* end-stage kidney disease; *GFR,* glomerular filtration rate; *mCrCl,* measured creatinine clearance; *mGFR,* measured glomerular filtration rate.

bacterial or fungal infection (this may be treated without affecting candidacy), 24-hour urine stone panel, cystoscopy, imaging to rule out a urinary tract malignancy, and a kidney biopsy to rule out underlying kidney disease (a thin basement membrane may not be a contraindication to donation).[38,39]

KIDNEY STONES

A renal calculus in the donor's remaining kidney may affect kidney function if it results in ureteral obstruction. Reassuringly, living kidney donors do not appear to have an increased risk of kidney stones requiring treatment with surgical intervention compared with healthy, matched nondonor controls (median follow-up, 8 years).[40] Evaluation of kidney stones in living kidney donor candidates includes a history from the candidate, laboratory investigations, including urine samples for evaluation of persistent microscopic hematuria, and renal imaging, such as computed tomography (CT). If suspected, further investigations may be performed, including parathyroid hormone measurements and 24-hour urine collections for metabolic testing. A history of previous stones does not necessarily rule out donation, particularly small, unilateral, nonrecurrent stones.[19] There is also the option to remove small kidney stones at the time of procurement prior to transplantation.[41]

HYPERURICEMIA, GOUT, AND METABOLIC BONE DISEASE

Compared with nondonor controls, living kidney donors have an increased risk of gout (3.4% vs. 2.0%) after a median of 8 years after donation.[42] This may be due to the reduced ability of a single kidney to excrete excess uric acid, a precursor to gout. Although a comprehensive gout assessment is not usually conducted for all candidates, a predonation serum urate level is frequently ordered, as well as other biochemical indicators of metabolic kidney disease, including inorganic phosphate, calcium, and parathyroid hormone levels. A living kidney donor nephrectomy may lower the concentrations of 1,25-dihydroxyvitamin D and phosphate and raise the concentration of parathyroid hormone, with no appreciable effect on the concentration of calcium. Whether these changes in bone mineral metabolism alter skeletal fracture risk in living kidney donors is unknown. To date, a single study of over 2000 living kidney donors (median age, 43 years) matched to a segment of the general population selected for good health found that after a median follow-up of 7 years (maximum, 18 years), the rate of fragility (osteoporotic) fractures was no higher in donors compared with nondonors.[43]

BLOOD PRESSURE

Sustained elevated blood pressure is a common cause of kidney disease and, conversely, kidney disease may accelerate the development of high blood pressure. Candidates with hypertension are eligible for donation only if their blood pressure can be controlled with antihypertensive medications and they are without end-organ damage related to their hypertension.[19] The systolic and/or diastolic blood pressure thresholds and the nature of the antihypertensive medications used (e.g., number of agents, class of drugs, dosage used) to disqualify a candidate may vary across programs and according to other candidate characteristics. Trained personnel should perform blood pressure measurements on at least two separate occasions. An ambulatory (e.g., 24-hour) blood pressure monitor may be used if hypertension is suspected. Donor candidates with hypertension that can be controlled to less than 140/90 mm Hg using one or two antihypertensive agents, and who do not have evidence of target organ damage, may be acceptable for donation. The decision to approve donation in persons with hypertension should be individualized based on their demographic and health profiles in relationship to the transplantation program's acceptance risk threshold.

METABOLIC AND LIFESTYLE RISK FACTORS

Obesity is a strong risk factor for diabetes mellitus, cardiovascular disease, and kidney disease.[44] Living donor nephrectomy is more difficult for patients with excess visceral fat, increasing the risk of perioperative complications, including infection, blood loss, and delayed wound healing.[45] Various body mass index (BMI) cut points have been reported in the literature as absolute or relative contraindications to donation. Elevated serum glucose levels or glucose intolerance are also strong risk factors for diabetes mellitus. Apart from personal and family history assessments of diabetes mellitus (childhood, adult-onset, gestational), glycosylated hemoglobin and serum and urinary glucose levels are typically measured early in the assessment of all candidates. Fasting glucose and oral glucose tolerance tests are recommended for high-risk candidates (e.g., high random glucose, positive family history). According to the 2017 KDIGO living kidney donor guideline, donor candidates with type 1 diabetes mellitus should not donate. The decision to approve donor candidates with prediabetes or type 2 diabetes mellitus should be individualized based on their demographic and health profiles in relationship to the transplantation program's acceptance threshold. Donor candidates with prediabetes and type 2 diabetes mellitus should be counseled that their condition may progress over time and may lead to end-organ complications.[19] Less evidence is available to comment on the influence of predonation lipid levels (e.g., cholesterol, triglycerides, high-density and low-density lipoproteins) and smoking on donor candidacy although, notably, smoking was a strong risk factor for ESKD in healthy persons.[29] Candidates should be educated and encouraged to modify their dietary and smoking habits, but eligibility based on these factors may vary across programs. Smoking should be considered as part of a comprehensive risk assessment.

SCREENING FOR TRANSMISSIBLE INFECTIONS

To reduce the risk of viral transmission from the donor to the recipient, the evaluation should include assessment of prior history of infections, recent travel history, and virology screens early in the evaluation—and again within the 2 to 4 weeks of donation to minimize the window of infection.[46] The 2017 KDIGO living kidney donor guideline recommends screening for human immunodeficiency virus, hepatitis B and C, Epstein-Barr virus, cytomegalovirus, syphilis, urinary tract infection, and other potential infections based on geography and environmental exposures.[19] If a donor candidate is found to have a potentially transmissible infection, the donor candidate, intended recipient, and transplantation team should weigh the risks and benefits of proceeding with donation and develop a management plan if the decision is made to proceed with donation.

CANCER SCREENING

All candidates should be up to date with local cancer screening guidelines according to age, gender, and family history. Donors with active cancer are generally not eligible to donate. Donors with a prior history of successfully treated cancer with a high risk of recurrence may be excluded from donation because antineoplastic agents may be nephrotoxic and transmission of cancer from the donor to the recipient can have serious consequences for the immunocompromised recipient.[47] Candidates with a prior history of cancer with a low risk of recurrence may be considered on an individual basis. In some cases, candidates with small renal tumors (high-grade Bosniak renal cysts [III or higher] or small [T1a] renal cell carcinoma curable by nephrectomy) may be acceptable for donation, and the donor and recipient provide consent for the cancer to be resected at the time of donor nephrectomy.[48,49]

GENETIC KIDNEY DISEASES

If the donor candidate is biologically related to the intended recipient, the cause of the recipient's kidney disease should be well understood before accepting the candidate. Candidates with a genetic kidney disease generally are not eligible to become donors. If a candidate has a family history of a genetic kidney disease, the candidate may be eligible to donate if the risk of developing kidney disease after donation is acceptably low, and the risks are discussed with the candidate. Genetic diseases that may be assessed during the donor candidate evaluation include autosomal dominant polycystic kidney disease, *APOL1*-related kidney disease, atypical hemolytic uremic syndrome, Alport syndrome, Fabry disease, familial focal segmental glomerulosclerosis, and autosomal dominant tubulointerstitial kidney disease. If a donor candidate is of sub-Saharan African ancestry, testing for *APOL1* renal risk variants may be offered.[50,51] The presence of two *APOL1* renal risk variants increases the lifetime chance of developing kidney failure, even in the absence of donation. The effects of kidney donation on this risk are unknown but are a topic of active research including in the national '*APOL1* Long-Term Outcomes' (APOLLO) and "Living Donor Extended Time Outcomes" (LETO) studies in the United States.

PREGNANCY

Although donation does not preclude future pregnancy and childbearing, candidates are not evaluated and/or do not donate while they are pregnant. A history of hypertensive disorders related to pregnancy (e.g., preeclampsia, gestational hypertension) increase the risk of developing kidney failure later in life, and the severity, timing, and frequency of these conditions should be considered before determining the potential donor's candidacy. The risk of adverse outcomes in subsequent pregnancies after kidney donation is increased, as discussed later (see "Maternal and Fetal Complications").

PSYCHOSOCIAL ASSESSMENT

The 2017 KDIGO living kidney donor guideline recommends that a psychosocial assessment be conducted for all donors (regardless of relationship with the intended recipient), in the absence of the intended recipient (to reliably assess voluntariness), and by a professional who is not involved with the care of the intended recipient. A thorough psychosocial assessment should minimize the incidence of poor psychosocial outcomes postdonation by careful selection or treatment (e.g., counseling).[52] Quality of life is generally positive postdonation, but there have been cases of regret, depression, and financial hardships.[53,54]

KIDNEY PAIRED DONATION AND INCOMPATIBLE LIVING DONOR TRANSPLANTS

Unintended ABO-incompatible transplants should be avoided, with duplicate blood typing of both the donor and intended recipient. If the intended recipient has anti-A antibodies and the donor candidate has A-type blood, the A subtype should be determined to establish compatibility instead of ruling it out at this stage (e.g., an A-type donor may donate to an O-type recipient if the donor is of the non-A1 subtype). Donor candidates and the intended recipient should also undergo allelotyping for human leukocyte antigen (HLA) types A, B, C, DP, DQ, and DR. The presence and strength of donor-specific anti-HLA antibodies should also be measured in the recipient; these are in addition to complement-dependent cytotoxicity or flow cytometry crossmatching and Luminex assays to determine the history of sensitization. If the intended recipient develops donor-specific, anti-HLA antibodies between the time of the initial crossmatch and donation (e.g., acquired by viral infections, pregnancy, or blood transfusion), a final crossmatch is required before donation.

For incompatible pairs, candidates should be informed of the availability, risks, and benefits of KPD, incompatibility management strategies, and waiting for a suitable deceased donor. The emergence of KPD, together with logistical advances (e.g., coordinating multiway exchanges, shipping organs rather than the donor), has increased the potential for KPD to substantially increase the number of living donor transplants performed.[55,56] Some programs even permit compatible donors to participate in KPD if they choose to wait for a living donor who is younger or has fewer donor-specific antibodies.[57]

DONOR NEPHRECTOMY APPROACHES

The surgical approach to donor nephrectomy is based on the surgeon's experience, as well as the donor's history, physical examination, renal anatomy and function, and personal preference. Following an uncomplicated donor nephrectomy, most donors can expect to spend 2 to 7 days in hospital and return to work in 4 to 9 weeks.[58]

LAPAROSCOPIC VERSUS OPEN DONOR NEPHRECTOMY

Laparoscopic donor nephrectomy has replaced the open technique as the standard of care, with the laparoscopic approach accounting for more than 90% of all living donor nephrectomies performed in the United States.[4,5] The conversion rate from laparoscopic to open donor nephrectomy ranges from 1% to 1.8%.[59] The average operative time for laparoscopic nephrectomy is longer compared with open nephrectomy (3–4 hours vs. 2–3 hours) and is associated with increased warm ischemia time (mean difference, 1.8 minutes; range, 2–17 minutes).[58,59] It is also associated with less perioperative blood loss (mean difference, −99.6 mL) and reduced postoperative pain and analgesic use.[58,59] Compared with open nephrectomy, the laparoscopic approach has other advantages, including shorter hospital stay (mean difference, −1.3 days), faster return to work (mean difference, −16.4 days), and improved cosmetic appearance due to the smaller incisions.[58] Thus, laparoscopic nephrectomy offers reduced surgical risk, pain, and cost, with no difference in ureteric complications, reoperations, delayed graft function, acute rejection, 1-year graft function, and early graft loss or 1-year graft loss.[59] Advances in surgical techniques, including the use of hand-assisted laparoscopic donor nephrectomy, aim to improve these donor experiences and outcomes. By doing so, potential donors, including older donors, may be more willing to undergo nephrectomy.[60]

LEFT VERSUS RIGHT DONOR NEPHRECTOMY

The laterality of donor nephrectomy may be influenced by anatomic or functional factors. Multiple renal arteries, unilaterally or bilaterally, may be technically more complicated or may increase the risk of dissection and bleeding. Structural abnormalities, such as the presence of renal cysts or calculi, may result in the removal of the affected kidney, with or without additional bench excision, leaving the donor with the unaffected kidney. Similarly, for donors with significant differences in split kidney function (>55% vs. <45%), the recommendation is to leave the donor the kidney with better function.[19]

Beyond these considerations, left nephrectomy is generally preferred because it is technically easier due to the longer left renal vein compared with the right.[19,61] A systematic review and meta-analysis has found that compared with right laparoscopic donor nephrectomy, there is less delayed graft function and thrombosis with left laparoscopic donor nephrectomy.[62] However, there was no significant difference in other surgical outcomes, including operative times, blood loss, warm ischemia time, and length of stay.

LIVING KIDNEY DONOR FOLLOW-UP

When the first living kidney donor, Ronald Herrick, completed his initial donor evaluation, he asked his surgeon whether the hospital would be responsible for his health care after donation.[1] The surgeon replied, "Ronald, do you think anyone in this room would ever refuse to take care of you if you needed help?" Since then, postdonation follow-up care continues to be an important consideration for living kidney donors.[63] A personalized plan for early- and long-term living kidney donor follow-up care should be discussed during the evaluation process.[19,64] This follow-up care plan may outline the timing and frequency of visits, as well as who will be providing the follow-up care (primary care provider vs. transplantation center). Early follow-up care (within 3–6 months) is routinely done, usually by the surgical team, as part of the postoperative care plan.[65]

The 2017 KDIGO living kidney donor guideline recommends lifelong annual assessment of kidney health, including blood pressure measurement, serum creatinine level testing for eGFR, and urine sample for albuminuria.[19] Currently, it is unclear how frequently living kidney donors are receiving follow-up care. In 2013, the United Network for Organ Sharing implemented mandatory reporting of clinical and laboratory data for living kidney donors at 6, 12, and 24 months after donation by the transplantation centers to the national transplant registry.[66] Although this led to improved rates of timely and complete donor follow-up data collection across the United States (from 33% prepolicy to 54% postpolicy), 43% of centers still did not meet the mandatory reporting threshold by 2015.[67,68] Donor follow-up data beyond 2 years, when medical conditions are more likely to develop, is not currently mandated in the United States.[69,70] Follow-up care at 1, 5, 10, and 15 years in Norway is achieved by 99%, 95%, 84%, and 77% of donors, respectively.[19] In a retrospective Canadian study of 534 living kidney donors from a single province, with a median follow-up of 6.5 years (maximum, 12.5 years), 25% of donors had all three markers of care—physician visit, serum creatinine level, albuminuria testing—performed on an annual basis.[71] The adherence to physician visits was higher than serum creatinine level or albuminuria testing (Fig. 71.4).

Unfortunately, there are barriers to follow-up for living kidney donors, including sociodemographic and geographic factors, perception of unnecessary need for follow-up due to excellent donor health, outdated donor contact information, cost of follow-up to the donor, and time constraints.[65,72–74] Donor factors associated with lack of follow-up includes younger age, black ethnicity, lack of insurance, lower educational attainment, and greater distance from the transplantation center.[67,75] Available resources and personnel may also limit monitoring by transplantation centers. Currently, there are no data to support the clinical and cost-effectiveness of lifelong follow-up of an otherwise healthy population. In addition to this, instead of providing reassurance of donor well-being, obligatory follow-up care may burden donors or give them an undue sense of concern about their health.[72]

However, there are numerous reasons to ensure adequate long-term follow-up care for living kidney donors. It allows the health care team to remain invested in the medical and psychosocial well-being of donors who have undergone an altruistic act for the benefit of another. Follow-up care may result in prevention, early detection, and treatment of diseases, particularly those associated with reduced kidney function and its complications. It also provides an opportunity to document medical and surgical complications that may be attributed to donation. This information can be used to develop interventions to mitigate these risks and can also improve the informed consent process for future living kidney donors. Follow-up care may also improve the living donor experience.[76] Lastly, ongoing follow-up care allows the health care team to update and educate prior donors on relevant outcomes as new research arises.

Recognizing the importance of living donor follow-up, the European Living Donation and Public Health project has mandated living donor registration and follow-up data collection through a centralized donor database.[77] In the United States, the Health Resources and Services Administration asked the Scientific Registry of Transplant Recipients (SRTR) to establish a national scientific registry for living donors, the Living Donor Collective.[78] This registry aims to collect follow-up data on living donors, including living donor candidates who undergo the evaluation process, but do not end up donating. Through staff contact and novel data linkages, the registry seeks to promote lifelong living donor follow-up.

Despite challenges in achieving it, most transplantation centers believe that living donor follow-up is a high priority and could mitigate adverse outcomes.[64] The transplant community has an obligation to promote the integrity, quality, and safety of the donation and transplantation system.[19] In doing so, transplantation programs may improve patient satisfaction and trust with the donation process, which could lead to increased rates of LDKT.

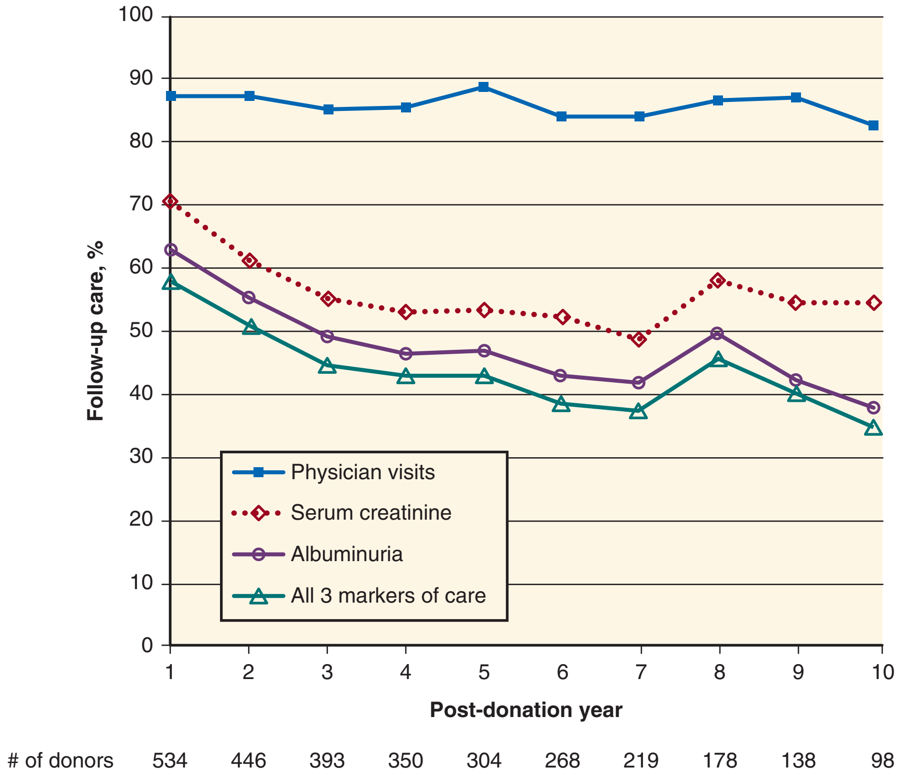

Fig. 71.4 Proportion of donors who have evidence of follow-up care during each postdonation year. (From Lam NN, Lentine KL, Hemmelgarn B, et al. Follow-up care of living kidney donors in Alberta, Canada. *Can J Kidney Health Dis.* 2018;5:1–11.)

RISKS AND OUTCOMES OF LIVING KIDNEY DONATION

Although most living kidney donors will not experience major morbidity or mortality after donation, donation does carry some risk of surgical, medical, and psychosocial complications.[69] Unfortunately, comprehensive data on long-term outcomes for living kidney donors are limited due to single-center studies with small sample sizes, insufficient power to quantify rare events, significant missing data or data lost to follow-up, short duration of follow-up, lack of donor diversity in race, ethnicity, and comorbidities, and lack of a comparable control group.[72,73,78–80] Living kidney donors are carefully selected following a rigorous evaluation process (see "Evaluation and Selection of the Living Kidney Donor Candidate") and, as a result are inherently healthier than the general population. Comparing outcomes in donors with nondonors selected for comparable baseline health offers a perspective of risk that can better capture the attributable risk of donation (Fig. 71.5).[19,80] In addition to this, the characteristics of living kidney donors have evolved over time. Transplantation programs are now accepting donors who are older, with more comorbidities, including hypertension and obesity.[81] Such practices are defensible if the risks, based on the candidate's demographic and health profiles, are acceptably low. Ongoing work is needed to capture and quantify risks in groups with less aggregate follow-up to date, such as ethnic minorities and medically complex donors.[72,81]

Clinical Relevance

Explicit recognition of perspectives of comparison is critical for drawing inferences about donor health outcomes (e.g., estimation of predonation risk, absolute postdonation risk, and donation-attributable risk). Recent studies grounded in comparison to healthy nondonor controls have identified small, donation-attributable increases in the long-term risks of kidney failure and gestational hypertension and preeclampsia in living kidney donors. These observations emphasize the need for careful donor candidate evaluation, selection, and informed consent. Donor candidates should have adequate time to consider the information provided to them during the evaluation process, although the duration of adequate time is not well defined and may vary according to donor characteristics. Ongoing efforts to develop best practices for the disclosure of donor risks and verification of comprehension are important research priorities.

PERIOPERATIVE MORTALITY AND COMPLICATIONS

In a study of 80,347 living kidney donors in the United States from 1994 to 2009, the 90-day surgical mortality rate was 3.1/10,000 donors.[82] This rate was higher in men than in women (5.1 vs. 1.7/10,000 donors; risk ratio, 3.0), in black

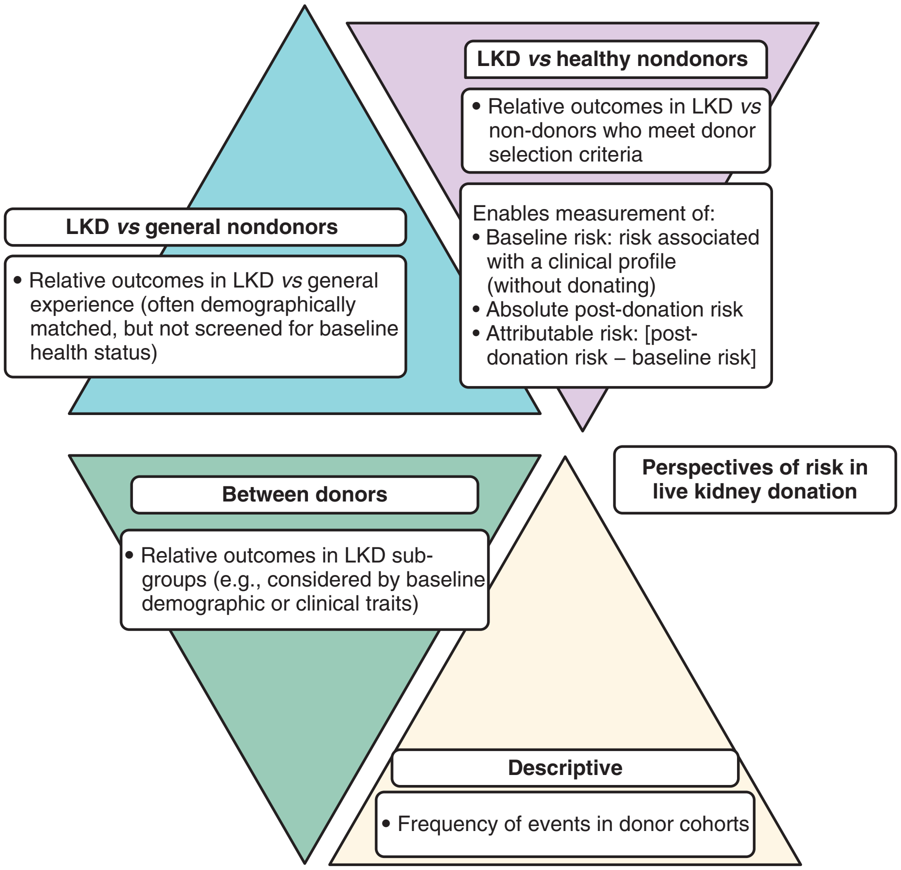

Fig. 71.5 Perspectives of risk in living kidney donation *(LKD).* (Modified from Lentine KL, Kasiske BL, Levey AS, et al. KDIGO clinical practice guideline on the evaluation and care of living kidney donors. *Transplantation*. 2017;101(8 Suppl 1):S7–S105; and Lentine KL, Segev DL. Understanding and communicating medical risks for living kidney donors: a matter of perspective. *J Am Soc Nephrol.* 2017;28(1):12–24.)

versus white and Hispanic donors (7.6 vs. 2.6 and 2.0/10,000 donors; risk ratio, 3.1), and in donors with hypertension versus those without (36.7 vs. 1.3/10,000 donors; risk ratio, 27.4). In comparison, the 90-day surgical mortality following donor nephrectomy was much lower compared with laparoscopic cholecystectomy (18/10,000 cases) and nondonor nephrectomy (260/10,000 cases).[83,84]

The incidence of perioperative morbidity has been reported to be 18% to 22% for minor complications and less than 3% to 6% for major complications.[80,81,85–88] A study integrating national donor registry data in the United States from 2008 to 2012 with administrative records from a consortium of 98 academic hospitals found that 16.8% of living kidney donors (n=14,964) experienced a perioperative complication, usually gastrointestinal (4.4%), bleeding (3.0%), respiratory (2.5%), and surgical and anesthesia-related injuries (2.4%).[85] Major complications, defined as Clavien severity level 4 or 5, were identified in 2.5% of donors. After adjustment for demographic, clinical (including comorbidities), procedural, and center factors, compared with white donors, black donors had significantly higher risks of experiencing any complication (18.2% vs. 15.5%) and of experiencing major complications (3.7% vs. 2.2%). Other significant correlates of major complications included obesity, predonation blood disorders, psychiatric conditions, and robotic nephrectomy, whereas greater annual hospital volume predicted lower risk. Another study has reported that procedure-related complications were similar for living donor nephrectomies, as they were for cholecystectomies or appendectomies, and all were significantly lower than nondonor nephrectomies.[81]

LONG-TERM MORTALITY

A meta-analysis of 19 studies involving 8098 living kidney donors has reported that the pooled mortality is 3.8%.[89] The risk of all-cause mortality is lower in living kidney donors compared with the general population, likely due to the rigorous evaluation and selection process.[90–93] When compared with age- and comorbidity-matched participants in the third National Health and Nutrition Examination Survey (NHANES), Segev and associates reported that the long-term mortality for living kidney donors in the United States, after a median follow-up of 6 years, was similar or lower, even when stratified by age, gender, and race (Table 71.4).[82] One study from Norway has reported that in the first decade of follow-up, the risk of all-cause mortality is similar in donors (n=1901) compared with healthy matched nondonors from the Health Study of Nord Trøndelag (n=32,621); however, at 25 years of follow-up, the cumulative incidence of all-cause mortality was higher in donors compared with nondonors (~18% vs. ~13%; adjusted hazard ratio [HR], 1.3; 95% confidence interval [CI], 1.1–1.5).[70] The most common cause

Table 71.4 Long-Term Medical Outcomes in Living Kidney Donors Compared With Selected Healthy Controls

Reference	Living Kidney Donors (n)	Healthy Matched Nondonors (n)	Median Donor Follow-up Time (years)	Donor Age at Donation (years)[a]	No. of Donor Events (%) [Event Rate]	No. of Healthy Nondonor Events (%) [Event Rate]	Event Rate Hazard Ratio (95% Confidence Interval)	*P* Value
End-Stage Kidney Disease								
Mjøen[70]	1901	32,621[b]	15.1	46 (11)	9 (0.47%)	22 (0.067%)	11.38 (4.37–29.63)	*P* < .001
Muzaale[115]	96,217	96,217	7.6	40 (11)	99 (0.10%) [30.8/10,000]	36 (0.037%) [3.9/10,000]	—	*P* < .001
Acute Kidney Injury Treated With Dialysis								
Lam[140]	2027	20,270	6.6	43 [34–50]	1 (0.05%) [6.5/100,000]	14 (0.07%) [9.4/100,000]	0.58 (0.08–4.47)	*P* = .61
All-Cause Mortality								
Mjøen[70]	1901	32,621	15.1	46 (11)	224 (11.8%)	2425 (7.4%)	1.30 (1.11–1.52)	*P* = .001
Segev[82]	80,347	80,347	6.3	—	–(1.5%)	–(2.9%)	—	*P* < .001
Reese[141]	3368	3368	7.8	59 (–)	115 (3.4%) [4.9/1000]	152 (4.5%) [5.6/1000]	0.90 (0.71–1.15)	*P* = .21
Cardiovascular Mortality								
Mjøen[70]	1901	32,621	15.1	46 (11)	68 (3.6%)	688 (2.1%)	1.40 (1.03–1.91)	*P* = .03
Death or Cardiovascular Event								
Reese[141]	1312	1312	—	59 (–)	—	—	1.02 (0.87–1.20)	*P* = .70
Garg[95]	2028	20,280	6.5	43 [34–50]	42 (2.1%) [2.8/1000]	652 (3.2%) [4.1/1000]	0.66 (0.48–0.90)	*P* = .008
Major Cardiovascular Events								
Garg[95]	2028	20,280	6.5	43 [34–50]	26 (1.3%) [1.7/1000]	287 (1.4%) [2.0/1000]	0.85 (0.57–1.27)	*P* = .43
Kidney Stones With Surgical Intervention								
Thomas[40]	2019	20,190	8.4	43 [34–50]	16 (0.79%) [8.3/10,000]	179 (0.89%) [9.7/10,000]	1.04 (0.60–1.80)	*P* = .90
Major Gastrointestinal Bleeding								
Thomas[142]	2009	20,090	8.4	42 [34–50]	33 (1.6%) [18.5/10,000]	253 (1.3%) [14.9/10,000]	1.25 (0.87–1.79)	*P* = .24
Skeletal Fractures								
Garg[143]	2015	20,150	6.6	43 [34–50]	[16.4/10,000]	[18.7/10,000]	0.87 (0.58–1.31)	*P* = .50
Gout								
Lam[42]	1988	19,880	8.4	43 [35–51]	67 (3.4%) [3.5/1000]	390 (2.0%) [2.1/1000]	1.6 (1.2–2.1)	*P* < .001
Gestational Hypertension or Preeclampsia								
Garg[130]	85[c]	510[c]	11.0	29 [26–32]	15 (11.5%)	38 (4.8%)	2.4 (1.2–5.0)[d]	*P* = .01

[a]Data presented as mean (standard deviation) or median [interquartile range].
[b]Living kidney donors were not matched to healthy nondonors in the comparison for end-stage kidney disease risk.
[c]In 85 donors, there were 131 pregnancies in follow-up. In 510 nondonors, there were 788 pregnancies in follow-up.
[d]Presented risk estimate is an odds ratio rather than a hazard ratio.

Modified from Lam NN, Lentine KL, Levey AS, et al. Long-term medical risks to the living kidney donor. *Nat Rev Nephrol.* 2015;11(7): 411–419.

of death among living kidney donors is cardiovascular disease, accounting for 30% to 40% of all deaths.[70,90] Although other studies have not shown a higher risk of cardiovascular mortality in donors compared with healthy matched nondonors, the Norwegian study also reported an increased risk in their donors (adjusted HR, 1.4).[70,94,95] Limitations of the Norwegian study included differences in baseline characteristics (e.g., age) and differences in accrual periods between donors and nondonors.[69,70,96,97]

KIDNEY FUNCTION

Despite losing 50% of functioning renal mass, living kidney donors are typically left with 70% of their predonation GFR due to compensatory hyperfiltration of the remaining kidney.[36,98–100] In the first decade following donation, 40% of donors have a GFR between 60 and 80 mL/min/1.73 m^2, 12% of donors have a GFR between 30 and 59 mL/min/1.73 m^2, and 0.2% of donors have a GFR less than 30 mL/min/1.73 m^2.[98] Following donation, there does not appear to be an accelerated loss of function in the remaining kidney, above that expected with normal aging.[100] A prospective study of 198 living kidney donors and 194 controls from the United States reported that the linear slope of the measured GFR by iohexol clearance, between 6 and 36 months of follow-up, increased among donors (+1.47 mL/min/year) and decreased among controls (−0.36 mL/min/year).[101] Several investigators have evaluated the change in renal functional reserve (RFR) capacity in donors postnephrectomy.[102] The value of this test in predicting long-term kidney function postdonation is unclear; however, RFR capacity appears to be blunted or lost in older and obese donors.[103]

END-STAGE KIDNEY DISEASE

Many transplantation programs and countries have reported their incidence of ESKD, typically defined as undergoing chronic dialysis or kidney transplantation.[91,104–114] A meta-analysis of 12 studies involving 108,900 living kidney donors has reported that the overall prevalence of ESKD is 0.5% in the 6 months to 5 years postdonation and 1.1% in the 10 years after that.[89] One study from Japan has reported the clinical course of eight living kidney donors who subsequently developed ESKD after a mean of 16 years.[108] For these donors, their renal function remained stable postdonation until they experienced an inciting event, such as infection, or a new comorbidity, such as hypertension or proteinuria, that led to progressive CKD.

The risk of ESKD in living kidney donors is lower than that in the general unscreened population (134 vs. 354 cases/million person-years).[114] In contrast, two studies have reported an increased risk of ESKD in living kidney donors compared with healthy matched nondonors.[70,96,115] In the previously described study from Norway, the incidence of ESKD over a median of 15 years was 0.5% (n = 9) for living kidney donors and 0.07% (n = 22) for healthy, matched nondonors (HR, 11.4; 95% CI, 4.4–29.6).[70] All affected donors were biologically related to their recipient. In another study of 96,217 living kidney donors from the United States, after a median follow-up of 7.6 years (maximum, 15.0 years), 99 donors (0.10%) developed ESKD after a mean of 8.6 years after donation.[115] The 15-year risk of ESKD was higher in donors who were men versus women, black versus white, and older versus younger (≥50 vs. <50 years). There was no significant difference between donors who were biologically related to their recipient versus those were unrelated to their recipient. In addition, the risk of ESKD was 86% higher among obese compared with nonobese donors (HR, 1.86; 95% CI,1.1–3.3); this risk increased by 7% for each unit increase in donor BMI above 27 kg/m^2 (HR 1.07; 95% CI,1.0–1.1).[116] Compared with healthy matched nondonors selected from the third NHANES study, the estimated 15-year risk of ESKD in this study was higher in living kidney donors (30.8 vs. 3.9/10,000; risk attributable to donation, 26.9/10,000).[19,115] The estimated lifetime risk of ESKD was also higher in donors compared with healthy, matched nondonors (90 vs. 14/10,000) but lower when compared with the unscreened general population (326/10,000). Despite these two studies, which reported a higher relative risk of ESKD in donors compared with healthy matched nondonors, the absolute risk of ESKD for living kidney donors is low. In a qualitative study of living kidney donors who subsequently developed ESKD, most donors did not regret their decision to donate, despite the adverse outcome.[117] Interestingly, recipients of kidneys from donors who developed ESKD also had a higher risk of graft loss (74% vs. 56%) and death (61% vs. 46%) at 20 years compared with those who received kidneys from donors who did not develop ESKD, suggesting some intrinsic risk conferred by the donor kidney.[118]

PROTEINURIA

In the general population, a higher level of proteinuria is associated with an increased risk of CKD, ESKD, cardiovascular disease, and death, independently of the GFR.[119] In a systematic review, the pooled incidence of postdonation proteinuria was 12%.[98] The degree of proteinuria was higher in donors (n = 129; 147 mg/day) compared with controls (n = 59; 83 mg/day) and increased with the time from donation.[98] In a study of 3956 white living kidney donors, the incidence of proteinuria after 17 years of follow-up was 6.1% and was higher in male donors (HR, 1.6) and in donors with an increased predonation BMI (HR, 1.1).[120]

HYPERTENSION

In addition to the increase in blood pressure that occurs with normal aging, the postdonation reduction in kidney function may contribute to a further increase in blood pressure.[121] A systematic review, which included 157 living kidney donors, has reported that after an average of 5 years, the blood pressure is 5 mm Hg higher in donors compared with nondonors (weighted mean for systolic blood pressure, 6 mm Hg; weighted mean for diastolic blood pressure, 4 mm Hg).[122] The risk of postdonation hypertension may be higher in black donors compared with white donors.[123,124] In a study of 3956 white living kidney donors, postdonation hypertension was associated with a more than twofold risk of ESKD.[120]

CARDIOVASCULAR EVENTS

In the general population, a reduced GFR is associated with cardiovascular events.[125–127] In living kidney donors who have undergone a surgical reduction in their GFR, the risk of long-term cardiovascular events does not appear to be increased compared with healthy matched nondonors.[95] In 2028 living kidney donors from Ontario, Canada (1992–2009), the risk of death-censored cardiovascular events (including myocardial infarction, stroke, and coronary or other revascularization) was similar to 20,280 healthy matched nondonors (1.7 vs. 2.0 events/1000 person-years; HR, 0.85; median follow-up, 7 years; maximum, 18 years). However, one study from Norway did report a higher risk of cardiovascular death in donors compared with nondonors (adjusted HR, 1.4; 95% CI,1.0–1.9).[70]

MATERNAL AND FETAL COMPLICATIONS

Compared with predonation pregnancies, postdonation pregnancies may be associated with a lower likelihood of full-term deliveries and a higher risk of fetal loss, gestational diabetes, gestational hypertension, proteinuria, and preeclampsia.[128,129] This risk may be attributed to other factors, such as maternal age at the time of pregnancy, rather than to donation. In a retrospective study of 85 female donors with 131 pregnancies from Ontario, Canada (1992–2009), the risk of gestational hypertension and preeclampsia was higher compared with 510 healthy matched nondonors with 788 pregnancies (11% vs. 5%; odds ratio, 2.4).[130] In this study, there were no significant differences in preterm births, cesarean section, postpartum hemorrhage, or low birth rate between donors and nondonors. Although most donors will have uncomplicated pregnancies and deliveries postdonation, it is recommended that pregnant donors have access to adequate perinatal care to reduce the risk of pregnancy complications.[19]

ETHICAL, LEGAL, AND POLICY CONSIDERATIONS

In 1988, Iran became the first and only country to implement a state-regulated, government-funded program to compensate living unrelated kidney donors financially.[131] This controversial program, known as the "Iranian model," has raised concerns about the exploitation of living kidney donors who may be from vulnerable populations, such as the undereducated and impoverished. Over 100 countries support the Declaration of Istanbul, which prohibits the practice of organ trafficking and transplant tourism, although it is estimated that 5% to 10% of kidney transplants globally are the result of commercial organ transplants.[132,133] Despite this, there is growing evidence that the out of pocket costs prevent potential candidates from completing the evaluation or donating.[134] Reimbursing the costs incurred by donor candidates or reimbursing the donor's funeral costs have been well-received by the public and other stakeholders as viable options for negating financial disincentives.[135]

Local laws and regulations on living donation should be followed and communicated clearly to donor candidates. Countries and programs should be explicit regarding their legal mandate to protect vulnerable populations, respect donor candidate autonomy (to be considered as candidates or withdraw at any time from the evaluation in a way that is confidential and protected), and prohibit incentives to donation that could create undue influence on the donor candidate's decision to donate. Countries and programs should also be clear on how donor candidates may be identified (e.g., living donor "champions," use of social media to advertise the need for a donor, or independent living donor advocates), as well as the policy on postdonation follow-up.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. Which of the following is not a risk of living kidney donation described in the literature?
   a. Death in the perioperative period (≈1 in 3300 donor nephrectomies)
   b. Postdonation gestational hypertension or preeclampsia in pregnancy (≈1 in 10 donors will develop outcome compared with 1 in 20 nondonors)
   c. A higher risk of fracture (10-year risk: ≈1 in 300 donors will fracture compared with 1 in 600 nondonors)
   d. A higher risk of end-stage kidney disease (40-year risk: 1 in 100 donors will develop kidney failure compared with 1 in 400 nondonors)

   Answer: c

   Rationale: As with any surgery, there is a perioperative risk of death with living kidney donation. Three studies to date have suggested a higher risk of gestational hypertension and preeclampsia in donor pregnancies compared with nondonor pregnancies. Kidney donation results in a decrease in the concentration of serum phosphate and plasma calcitriol and an increase in the concentration of serum parathyroid hormone, with no change in the concentration of serum calcium. However, no study has shown that donors have a higher risk of fractures compared with nondonors (see Table 71.4). Two studies have highlighted a higher risk of kidney failure in donors compared with nondonors, but the absolute chance of developing this outcome in the 15 years after donation is low (<0.5%).

2. Which of the following is a standard test done in most transplantation centers for candidates who are evaluated to become a living kidney donor?
   a. Native kidney biopsy
   b. Hepatitis B serology
   c. 24-Hour urine collection for kidney stone metabolic testing
   d. 24-Hour ambulatory blood pressure
   e. Oral glucose tolerance test

   Answer: b

   Rationale: To prevent unintended transmission of infection from a living donor to a recipient, the donor candidate has recommended tests for infectious disease screening. Some of these tests are also repeated very close to the transplant surgery. The 2017 KDIGO guideline recommends screening for the following viruses: human immunodeficiency virus, hepatitis B and C, Epstein-Barr virus, cytomegalovirus, and syphilis. The regulatory agencies of many countries also have specific guidance concerning infectious disease testing in living donors. All the other tests listed are done in some (but not all) donor candidates based on their risk factors and initial test results.

3. Which of the following is not a goal of the living kidney donor candidate evaluation?
   a. Educate donor candidates about the process and likely outcomes of living kidney donor transplantation.
   b. Determine whether the candidate has acceptable medical and psychological risks of donation to proceed with nephrectomy.
   c. Minimize the medical and psychological risks of donation prior to nephrectomy for candidates who are proceeding with nephrectomy.
   d. Develop a postdonation plan to minimize the short- and long-term medical and psychosocial risks of donation.
   e. Ensure that the intended recipient maintains a good relationship with the donor candidate after the donation process.

   Answer: e

   Rationale: The 2017 KDIGO guidelines describe the goals of the evaluation of living kidney donor candidates. This includes understanding a donor candidate's willingness to donate a kidney voluntarily without pressure, assessing the benefits and risks of donation for each donor candidate, supporting candidate decision making through education and counseling, developing a plan for postdonation follow-up and care and, for excluded donor candidates, developing a plan for any needed care and support. The evaluation team can provide advice to donors, including what might happen to their relationship with the recipient after donation. However, the evaluation team is not in a position to ensure that recipients maintain a good relationship with donors after the donation process.